

Question	Answer	Mark	Mark scheme	Additional guidance
24	$4(2 + \sqrt{2})a^2$	P1	for process to find area required and identifying 135° or 45° or 90° angle(s), eg splitting shape into square and 4 triangles and an angle relevant to the triangle clearly identified.	90° must be in a triangle to gain credit. May be seen on diagram.
		P1	for process to find the area of a relevant triangle using 45°, eg $\frac{1}{2} \times a \times \left(2 \times \frac{a}{\sqrt{2}}\right) \times \frac{1}{\sqrt{2}} \left(= \frac{a^2}{2}\right)$ or using 90°, eg $\frac{1}{2} \times a \times a \left(= \frac{a^2}{2}\right)$ or process to find the area of a square made from 2 small triangles, eg $a \times a \left(= a^2\right)$	Accept $0.49 \dots a^2$ May be seen as the area of 2 squares (from 4 small triangles)
		P1	for process to find the length of the square, eg $a + a + \sqrt{a^2 + a^2} \left(= 2a + a\sqrt{2}\right)$	Accept $3.41a$
		P1	for process to find the total area, eg $\left(2a + a\sqrt{2}\right)^2 + 4 \times \frac{a^2}{2}$	Accept $(11.655 + 4 \times 0.49) a^2$
		A1	(dep on P3) for $4(2 + \sqrt{2})a^2$	Answer only award no marks. If working in decimals accept $\frac{13.656}{2 + \sqrt{2}}$ leading to 4 Accept $p = 4$ if supported by correct working